PERFECT COMPLEXES ON NOETHERIAN SCHEMES INTERACT COHERENTLY WITH THE COHERENT SUBCATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. The source ends 'Some details omitted'; the omitted step is the per-case degree bookkeeping for parts (2)-(5), carried as an explicit obligation.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 2967 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Local freeness of perfect complexes on quasi-compact spaces. Cohomology, lemma perfect-on-locally-ringed (cross-chapter, named). E is represented by a bounded complex of finite free modules locally.
- Complex-level computations of tensor and internal Hom. Cohomology, lemmas Rhom-strictly-perfect and bounded-flat-K-flat (cross-chapter, named). Both E tensor^L L and R SheafHom(E, L) are computed by the finite-free complex E-bullet.
- Cohomology sheaves of complex-level constructions. structural. Finiteness/boundedness of H^i is read off the finite-free complex directly.

Source reference. Original reference 08DM: focused-seeds/geometry/S9-perf2-perfect-on-noetherian/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06YD: focused-seeds/geometry/S9-perf2-perfect-on-noetherian/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BCJ: focused-seeds/geometry/S9-perf2-perfect-on-noetherian/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a Noetherian scheme. Let E in $D(\mathcal{O}_X)$ be perfect. Then*

(1) *E is in $D^b_{Coh}(\mathcal{O}_X)$,*

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(2) *if* L *is in* $D_{Coh}(\mathcal{O}_X)$ *then* $E \otimes^{\mathbf{L}}_{\mathcal{O}_X} L$ *and* $R\mathcal{H}om_{\mathcal{O}_X}(E, L)$ *are in* $D_{Coh}(\mathcal{O}_X)$,
(3) *if* L *is in* $D^b_{Coh}(\mathcal{O}_X)$ *then* $E \otimes^{\mathbf{L}}_{\mathcal{O}_X} L$ *and* $R\mathcal{H}om_{\mathcal{O}_X}(E, L)$ *are in* $D^b_{Coh}(\mathcal{O}_X)$,
(4) *if* L *is in* $D^+_{Coh}(\mathcal{O}_X)$ *then* $E \otimes^{\mathbf{L}}_{\mathcal{O}_X} L$ *and* $R\mathcal{H}om_{\mathcal{O}_X}(E, L)$ *are in* $D^+_{Coh}(\mathcal{O}_X)$,
(5) *if* L *is in* $D^-_{Coh}(\mathcal{O}_X)$ *then* $E \otimes^{\mathbf{L}}_{\mathcal{O}_X} L$ *and* $R\mathcal{H}om_{\mathcal{O}_X}(E, L)$ *are in* $D^-_{Coh}(\mathcal{O}_X)$.

Proof. Since X is quasi-compact, each of these statements can be checked over the members of any open covering of X. Thus we may assume E is represented by a bounded complex $\mathcal{E}^\bullet$ of finite free modules, see Cohomology, Lemma 0BCJ. In this case each of the statements is clear as both $R\mathcal{H}om_{\mathcal{O}_X}(E, L)$ and $E \otimes^{\mathbf{L}}_{\mathcal{O}_X} L$ can be computed on the level of complexes using $\mathcal{E}^\bullet$, see Cohomology, Lemmas 08DM and 06YD. Some details omitted. □